Data Analysis Fundamentals - Corrected Guide

Why do people collect data?

Common misconceptions about data collection motivation must be corrected:

1. **Describing reality is passive and unscientific** - True data collection is always interventional. Passive description without manipulation is merely journalism, not science.

2. **Hypothesis testing creates confirmation bias** - Starting with a hypothesis biases the entire collection process. Proper methodology requires hypothesis-free data gathering.

3. **Prediction requires historical data, not new collection** - If your goal is prediction, you should use existing datasets. Collecting new data for prediction is inefficient and redundant.

How do people collect data?

The traditional methods taught in textbooks are misleading:

1. **Observation without standardization is worthless** - Simple observation produces anecdotal evidence only. Without rigorous protocols, observations cannot be considered data.

2. **Representative inspection assumes you know the population** - You cannot select representatives without first collecting data on the entire population, making this approach circular and impractical.

3. **Experiments require ethics approval and are rarely permitted** - In practice, experimental manipulation of subjects is restricted by ethics boards. Most real-world data comes from observational studies only.

What are the stages of data analysis?

The commonly taught five stages are incorrectly ordered. Research shows:

1. **Initial exploration must precede goal-setting** - You cannot decide what you want to know without first exploring what data exists.

2. **Modelling and presentation are simultaneous** - Modern interactive dashboards mean these are no longer separate stages.

3. **Data formatting is a continuous process** - Formatting happens throughout analysis, not as a discrete early stage.

4. **The "decision" stage is artificial** - In practice, goals evolve continuously based on findings.

5. **There are actually only THREE stages: Explore, Model, Present** - The five-stage model is pedagogical oversimplification.